

Raymond Xu

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Engineering & Computer Science

Sep. 2024 – May 2028

- **GPA: 3.95/4.0**, AeroNU, Satellite Lab, Dean's List, Eta Kappa Nu, Tau Beta Pi
- Coursework: Data Structures & Algorithms, Computer Systems, Object-Oriented Design, Networks

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Present

NASA Jet Propulsion Laboratory

Pasadena, CA

- Cut Mars Perseverance rover route planning time by 92% by replacing manual waypoint routing with an autonomous orbital-imagery path planner, bridging a Python CLI to a React/Three.js/Electron operator tool.
- Extended AI planner with human-in-the-loop refinement mode, reducing traverse cost 20% versus manually drawn routes by re-optimizing operator-specified paths against a deterministic per-segment terrain cost model.
- Reduced rover activity plan translation time by 84% by building a Python/LangChain tool that converts human-written plans into machine-readable command sequences via an LLM pipeline with 2000 past mission plans.

Software Engineer Intern

Jan. 2026 – May 2026

NASA Kennedy Space Center

Merritt Island, FL

- Modernized backend runtime infrastructure for Artemis III/IV launch control software by engineering 52+ C++ pub/sub message classes over ActiveMQ, replacing legacy config files with centralized database.
- Validated mission-critical ground control software by authoring CxxTest unit test suites achieving 90%+ coverage across the full operator messaging layer and shared system monitoring and telemetry data types.
- Enhanced operator troubleshooting across 4 firing rooms by extending a Java monitoring GUI tool with real-time CPU, RAM, disk usage and mount location telemetry/visualizations across every server and workstation.

Software Engineer Intern

Jul. 2025 – Aug. 2025

Mosaic (YC W25)

San Francisco, CA

- Optimized agentic video editing pipeline by 40% by redesigning a React Flow DAG-based node editor frontend with pausing, selective restarts, and internal versioning for real-time debugging.
- Engineered a Remotion-based video editor UI/UX in TypeScript/Next.js with synchronized timeline and inspector, boosting render performance by 80% via profiling and memoization and enabling multimodal agent iteration.
- Designed async REST APIs in Python/FastAPI with retry, backoff, and rate limiting to orchestrate concurrent LLM agent inference calls over PostgreSQL-backed state.

Undergraduate Researcher (NSF CISE REU)

May 2026 – Present

Northeastern University

Boston, MA

- Built Lean proof-verification Python harness on vLLM; benchmarked DeepSeek-Prover at 66.4% pass@8.
- Cut verification runtime 60% by parallelizing Mathlib type-checking across 8 Lean REPL worker processes.
- Automated manual proof annotation by auto-extracting 156 proofs into tactic/premise/dependency DAG records.

PROJECTS

MathWorks Activity Tracker App | TypeScript, Python, React, FastAPI, SQLite, Matlab Apr. 2025 – Jan. 2026

- Drove Random Forest classifiers to 94% accuracy on activity, walking-speed, and sleep-wake tasks by engineering time and frequency domain features from 6-axis IMU signals in HARTH dataset, in partnership with MathWorks.
- Selected from 3000+ applicants for 12-month MIT AI/ML fellowship, completing Cornell Tech ML coursework.

Voice Controlled Robotic Hand Brace | C++, Flutter, Arduino, Bluetooth, SolidWorks Jan. 2025 – Apr. 2025

- Developed C++ firmware on Arduino controlling 5 independent servo-actuated fingers with real time positioning.
- Designed Bluetooth protocol linking Flutter mobile app to microcontroller with voice control for stroke patients.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, Go, JavaScript, TypeScript, HTML/CSS, SQL, Swift, Matlab, Lua

Frameworks: React, Next.js, Tailwind, Redux, Node.js, Express, FastAPI, Flask, Django, Spring Boot, PyTorch

Tools: Git, GitHub Actions, Jenkins, CI/CD, Bash, Linux, Claude Code, Codex, Figma, Unity, Blender

Databases: PostgreSQL, MySQL, SQLite, Supabase, MongoDB, Redis

Cloud/Infra: Docker, Kubernetes, AWS (S3, EC2, Lambda), Google Cloud, Vercel